

MUHAMMED AFSAL P M

Associate Software Engineer (AI)

Kochi, Kerala, India — Valid Indian Passport

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[Kaggle](#) — [HuggingFace](#)

PROFILE SUMMARY

Results-driven AI Engineer with 2+ years of experience specializing in agentic AI systems, multimodal conversational platforms, and production LLM pipelines across BFSI, healthcare, and enterprise automation. Expert in designing and deploying scalable AI solutions using Retrieval-Augmented Generation (RAG), multi-agent orchestration, FastAPI microservices, vector databases, and cloud-native deployments on AWS and Azure. Proven track record in building production-grade real-time voice and chat AI systems, integrating LLMs, embeddings, and distributed backend architectures to enable intelligent automation, decision support, and conversational workflows. Experienced in developing end-to-end machine learning systems including model training, fine-tuning, deployment, and monitoring with focus on performance, scalability, and reliability.

SKILLS SUMMARY

- **AI & ML:** LLMs, Generative AI, Machine Learning, RAG, LLM fine-tuning, Hugging Face Transformers, Embeddings, Model Deployment
- **Programming & Databases:** Python, SQL, FastAPI, MongoDB, MySQL, ChromaDB, FAISS, Qdrant, SQLite, Redis
- **Cloud & DevOps:** AWS (S3), Azure, Docker, Kubernetes, RabbitMQ, CI/CD, Git, SonarQube, Celery
- **Frameworks & Tools:** CrewAI, Streamlit, TensorFlow, scikit-learn, Power BI, LiveKit, Twilio
- **Other Skills:** Microservices Architecture, Data Engineering, Data Visualization, REST API Design, Async Programming, Prompt Engineering

WORK EXPERIENCE

Techvantage Systems Pvt. Ltd techvantage.ai

Dec 2024 – Apr 2026

Associate Software Engineer (AI)

- Designed and deployed multi-agent AI systems using CrewAI and LLMs (GPT-4, Groq, Llama), including LLM fine-tuning, to automate verification, risk assessment, and decision-making in banking, insurance, and audit domains.
- Engineered end-to-end ML pipelines following the FTI architecture with scikit-learn and MongoDB, automating feature extraction, model training, and inference with integrated monitoring.
- Built enterprise AI platforms for recruitment, contract intelligence, and HR policy, leveraging adaptive testing and LLM question generation to deliver real-time analytics and ensure compliance.
- Developed advanced computer vision and data retrieval systems using YOLO, DeepSORT, and ChromaDB, enabling automated document analysis, object tracking, and contextual report generation.
- Delivered production-grade solutions by integrating FastAPI, AWS, and SQL/NoSQL databases, implementing features like secure OTP authentication, multi-language RAG, fraud detection, and automated report generation.

Zentis AI (Startup incubated within Techvantage)

May 2025 – Oct 2025

Core Team Member – Associate Software Engineer (AI)

- Architected and developed a modular microservices-based AI platform from inception using autonomous agents, enabling workflow automation, document intelligence, and decision support across BFSI, healthcare, and manufacturing domains.
- Implemented AI orchestration pipelines using LLMs (Azure OpenAI, Groq), Retrieval-Augmented

Generation (RAG), vector search, and CrewAI multi-agent frameworks to support contextual reasoning and knowledge-driven automation.

- Built scalable backend services with FastAPI, MongoDB, and RabbitMQ; containerized with Docker and deployed on Kubernetes across Azure and AWS with secure APIs, fault tolerance, and distributed processing.
- Established production DevOps practices including CI/CD pipelines, automated testing, SonarQube code quality checks, monitoring, and compliance controls, improving system reliability, observability, and reducing manual audit effort.

Go-Do (*Startup incubated within Techvantage*)

Dec 2025 – Feb 2026

Core Team Member – Associate Software Engineer (AI)

- Developed a multilingual agentic AI concierge accessible via WhatsApp and voice interfaces, enabling users to complete real-world tasks through natural language commands across multiple Indian languages.
- Architected end-to-end conversational pipelines using FastAPI, LiveKit, Twilio WhatsApp API, Redis, MongoDB, and FAISS-based RAG, supporting real-time task orchestration, session persistence, and knowledge retrieval.
- Implemented multimodal AI workflows integrating STT, TTS, and LLM reasoning (Gemini, OpenAI, Sarvam), delivering scalable voice-and-text agents with tool-calling, analytics logging, and production-ready deployment.

Bharat Petroleum Corporation Limited, Kochi Refinery

Dec 2023 – Dec 2024

AI Engineer (Apprenticeship)

- Extracted, validated, and managed data from multiple sources using SQL/MySQL, ensuring high data quality and integrity for petroleum operations; enhanced workflows using agent-based AI for automated data validation and anomaly detection
- Conducted advanced data analysis using SQL, stochastic methods, and statistical modeling to optimize extraction, refining, and distribution processes, incorporating AI-driven insights for improved decision-making
- Developed and optimized machine learning models to support AI-powered solutions, including an AI chatbot integrated with a Retrieval-Augmented Generation (RAG) pipeline for context-aware responses
- Designed and implemented AI-driven data auditing and report generation systems using agent-based architectures and RAG to automate validation, summarization, and compliance reporting
- Built interactive dashboards with Power BI, Tableau, and Excel to enhance operational efficiency and data-driven decision-making, integrating AI-generated insights and predictive analytics

PROJECTS

- **Zentis.ai – Agentic AI Platform:** Built an enterprise-grade AI orchestration platform leveraging autonomous agents and scalable microservices to automate complex workflows across BFSI, healthcare, and compliance domains.
- **Go-Do – Multilingual Agentic AI Concierge:** Developed a production-ready conversational AI platform using multimodal agents, RAG, and tool-driven workflows to enable voice and WhatsApp-based task automation, multilingual interactions, and real-time service execution.
- **Publisher Website Classifier – LLM-Powered Publisher Vetting System:** Built an LLM-powered publisher vetting microservice with RAG (ChromaDB + sentence-transformers), dual-provider architecture (OpenAI/Ollama), and tiered web scraping to automate affiliate quality assessment.
- **AI Agents for Banking Loan Processing:** Developed AI agents to automate loan verification, eligibility checks, and decision-making workflows.
- **AI-Powered Recruitment Platform:** Built recruitment automation integrated with MongoDB for resume shortlisting, test scheduling, AI-generated questions, and candidate analytics.

- **Prediction Models:** Built ML models for cancer prediction, telecom churn prediction (AdaBoost, Gradient Boosting, Logistic Regression, Decision Tree), and price prediction using regression algorithms.
- **AI Chatbot:** Developed an AI-powered chatbot for HR and operational workflows using autonomous agents based intent classification, and LLMs for contextual responses.
- **Contract Management Tool:** Developed an AI-powered contract analysis and management system using FastAPI, LLMs (Groq/Local), ChromaDB, and MongoDB for automated compliance checks, risk assessment, clause detection, and intelligent contract Q&A chatbot with multi-format document support.

EDUCATION

Certified Specialist in Data Science and Analytics

May 2024 – Oct 2024

ICT Academy of Kerala

Bachelor of Technology in Computer Science and Engineering

Jul 2019 – Aug 2023

with minor in Electronics and Communication Engineering

First Class with Distinction | CGPA: 8.46

APJ Abdul Kalam Technological University

CERTIFICATIONS & PUBLICATION

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|---|-------------|-----------------|
| • Microsoft Certified: Azure Data Scientist Associate (DP-100) | Certificate | <i>Jun 2024</i> |
| • Microsoft Certified: Azure Data Fundamentals (DP-900) | Certificate | <i>Mar 2024</i> |
| • IBM Data Science Professional Certificate | Certificate | <i>Dec 2024</i> |
| • Introducing Multimodal Llama 3.2 | Certificate | <i>Jan 2025</i> |
| • Machine Learning Fundamentals Micro-Credential by Alteryx | Certificate | <i>Apr 2024</i> |
| • Generative AI Fundamentals by Databricks | Certificate | <i>Oct 2025</i> |
| • Publication: “Smart Application for Visually Impaired People,” IJARIE, Vol-9, Issue-3, 2023, ISSN(O) 2395-4396 | Publication | <i>Jun 2023</i> |